Setup Guide: Maze Solver

Time Allocation: 90 Minutes

Challenge Overview

The robot must autonomously navigate from the entrance to the exit of a corrugated plastic classroom maze.

Environment & Arena Setup

• **Material Preparation:** Acquire sheets of corrugated plastic (Coroplast) and cutting tools to build walls approximately 15 to 20 cm high. This height ensures line-of-sight for most standard classroom ultrasonic, infrared, or color sensors.

• **Constructing the Maze:** Build a single-path or mildly branched maze on a flat, high-friction floor (like low-pile carpet or matte vinyl flooring) to prevent wheel slippage. Ensure corridor widths are at least 30 to 40 cm to accommodate standard classroom robot chassis.

• **Marking Start and Finish:** Clearly label the 'Entrance' and 'Exit' using contrasting tape on the floor. If using light sensors for navigation, ensure the floor under the maze provides a consistent, non-reflective background (e.g., standard grey or white flooring).

• **Lighting Control:** Minimize dramatic shadows or direct sunlight streaming into the maze area, as ambient light fluctuations can interfere with infrared and light sensors.

• **Testing Area:** Set up a small testing station outside the main arena so students can debug basic movement and sensor readings before placing their robots into the official maze.